MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6922

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Twin triode, miniature, medium Mu

Outline --- 6-2 (EIA)

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	2a	2g	2k	h	h	1a	1g	1k	sd

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ebb	Eb	Ecc	Ec	Ehk	Rk/k
Unit:	V	V dc	V dc	V dc	V dc	v	Ohms
Maximum:	6.6	---	250	---	---	70	---
Minimum:	6.0	---	---	---	-110	-135	---
TEST CONDITIONS:	6.3	100	90 (approx)	+9	---	---	680

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Rg/g	Ik/k	Ic/g	Pp/a	TE	Alt
Unit:	Meg	mA dc	mA dc	W	°C	ft
Maximum:	0.5	22	---	1.65	170	(See note 1)
Minimum:	---	---	---	(See note 2)	---	---
TEST CONDITIONS:	--- (See note 3)	---	---	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	285	315	mA
1336	Heater-cathode leakage	4		0.4	II	Ihk	---	6	µA dc
1266	Total grid current (1)	4	Ebb = 90 V dc; Ecc/g = -1.5 V dc; Rk/k = 0	0.4	II	Ic	0	-0.1	µA dc
1256	Electrode current (1) (anode)	4	Ebb = 91 V dc; Ecc = 0; Rk = 80 ohms	0.4	II	Ib	11.6	18.9	mA dc
1256	Electrode current (2) (anode)	4	Ec = -15.0 V; Eb = 150 V	0.4	II	Ib	---	5	µA dc
1306	Transconductance (1)	4		0.4	II	Sm	10,000	15,000	µmhos
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	4	10 meg in series	2.5	I	---	---	---	---
1306	Transconductance (2)	4	Ef = 5.7 V ac	2.5	I	ΔSm Ef	---	15	%
1316	Amplification factor	4		6.5	I	Mu	26.5	39.5	---
1266	Total grid current (2)	4,5		2.5	I	Ic	0	-0.2	µA dc
2201	Noise and microphonics	6	Ebb = 250 Vdc; Rk = 340 ohms; Ecal = 100 mV ac; Rp = 5,000 ohms	2.5	I	---	---	---	---
1331	Direct-interelectrode capacitance	4	No shield	6.5	Code E	Cga	---	1.6	pF
		4	No shield			Cin	2.7	3.9	pF
			No shield (sec 1)			Cout	1.55	1.95	pF
			No shield (sec 2)			Cout	1.45	1.85	pF
			No shield			Cgg	---	0.006	pF
			No shield			Caa	---	0.050	pF
1031	Low-frequency vibration	6	Rp = 2,000 ohms	6.5	Code H	Ep	---	100	mV ac
1041	Shock	7	450 G	6.5	See note 8	---	---	---	---

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 2 -Continued								
1031	Vibration-fatigue		F = 50	6.5	See note 8	---	---	---	---
---	Shock and vibration-fatigue-test end points								
1031	Low-frequency vibration	6	Rp = 2,000 ohms	---	---	Ep	---	150	mV ac
1336	Heater-cathode leakage	4		---	---	Ihk	---	9	µA dc
1306	Transconductance (1)	4		---	---	Sm	8,500	16,500	µmhos
1266	Total grid current (1)	4	Ebb = 90 V dc; Ecc/g = -1.5 V dc; Rk/k = 0	---	---	Ic	---	-0.2	µA dc
1121	Base strain			---	---	---	---	---	---
2126	Envelope strain			4.0	I	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3						
1506	Heater-cycling life	9	Ef = 7.0 V; Ehk = +100 V dc; Ec = Eb = 0; 1 minute "on", 4 minutes "off"	---	---	---	---
---	Heater-cycling life-test end point						
1336	Heater-cathode leakage	4		Ihk	---	8	µA dc
1516	Stability life	9	Ehk = +70 V dc; Rg = 47,000 ohms; TA = room	---	---	---	---
---	Stability life-test end point						
1306	Change in transconductance (1) of individual tubes	4		ΔSm t	---	10	%
1501	Intermittent life	10	Ehk = +70 V dc; Rg = 47,000 ohms	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3 -Continued						
---	Intermittent life-test end points (1,000 hours)	11					
1266	Total grid current (1)	4	Ebb = 90 V dc; Ecc/g = -1.5 V dc; Rk/k = 0	Ic	0	-0.9	µA dc
1301	Heater current			If	285	315	mA
1306	Change in transconductance (1) of individual tubes	4		ΔSmt	---	25	%
1336	Heater-cathode leakage	4		Ihk	---	8	µA dc
1211	Insulation of electrodes	4		R	50	---	meg

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. The total absolute-maximum anode dissipation of both sections is 2.2 watts.
3. This value is for operation under fixed bias conditions. With cathode bias, Rg may be 1 megohm maximum.
4. Test each unit separately.
5. Prior to this test, tubes shall be preheated a minimum of 5 minutes with both sections operating at the conditions indicated below. The 3-minute test is not permitted. Test at preheat conditions within 3 seconds after preheating. Total grid current (2) shall be the last test performed on the sample selected for the total grid current (2) test.

Ef	Ecc	Ebb	Rk	Rg
V	V dc	V dc	Ohms	Ohms
7.0	+9	100	680	47,000

6. Tie 1k to 2k, 1g to 2g, and 1a to 2a. Parasitic suppressors of 50 ohms are permitted.
7. A grid resistor of 0.1 megohm shall be added, except when a thyratron-type short indicator is used.
8. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code E, shall apply.
9. The AQL shall be 1.0, inspection level S3.
10. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.
11. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.
12. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Air Force - 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-3301)

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